Chapter 6 Notes

Section 1

Notes

Note that $F \upharpoonright A$ does not necessarily equal $A \times F''(A)$. In fact, if $F''(A)$ contains more than one element (say it contains at least the two elements b, c), then obviously $F \upharpoonright A \neq A \times F''(A)$, as, for any element $x \in A$, we would have $\langle x, b\rangle$ and $\langle x, c\rangle$ both in $A \times F''(A)$, but there can only be one y for which $\langle x, y\rangle \in F \upharpoonright A$.

It is remarked that:"If ϕ_1 and ϕ_2 are functions from a class A onto a class B and if $\phi_1\phi_2^{-1}$ is an identity function, then obviously $\phi_1 = \phi_2$." Here is why this is. Now, note that this assumes that ϕ_2^{-1}, is, in fact, a function. Then, for every element y in the range of ϕ_2, we have $\phi_2^{-1}(y) = x$ for some x in the domain of ϕ_2, and $\phi_1\phi_2^{-1}(y) = \phi_1(x) = y$. But also, $\phi_2(x) = y$. Thus $\phi_1(x) = \phi_2(x)$ for every x in the domain of ϕ_2, and because they have the same domains, the functions are the same.

Note that if a class A is well ordered under $\subseteq$, then it is also properly well ordered under $\subseteq$. The proof of this is the same as the proof of Proposition 1.3.

Exercise 1.1

Let $\mathcal{N}$ be a nest of functions.

(1): We first show that $\cup\mathcal{N}$ is function. Obviously $\cup\mathcal{N}$ is a relation. Now, suppose we had $\langle x, y\rangle$ and $\langle x, z\rangle$ both in $\cup\mathcal{N}$ for some $y \neq z$. Then x is in the domain of some function f. Without loss of generality, suppose $\langle x, y\rangle \in f$. Then $\langle x, z\rangle \notin f$. Now, this would mean that $\langle x, z\rangle$ was in some other function g. But, because $\mathcal{N}$ is a nest, either $f \subseteq g$ or $g \subseteq f$. Without loss of generality, suppose $f \subseteq g$. Then we would have $g(x) = y$ and $g(x) = z$, a contradiction to g being a function. Hence $\cup\mathcal{N}$ must be a function.

(2): We now show that the domain of $\cup\mathcal{N}N$ is the union of the class of domains of the elements of $\mathcal{N}$ (call this D). Suppose $x \in \mathrm{Dom}(\cup\mathcal{N})$. Then, for some y and some function f, $\langle x, y\rangle \in f$, meaning $x \in D$. On the other hand, suppose $x \in D$. Then, for some y and some function f, $langlex, y\rangle \in f$, meaning $x \in \mathrm{Dom}(\cup\mathcal{N})$. Thus $\mathrm{Dom}(\cup\mathcal{N}) = D$. Similar reasoning shows $\mathrm{Ran}(\cup\mathcal{N})$ is the union of the class of ranges of the elements of $\mathcal{N}$.

(3): Lastly, suppose that every element of $\mathcal{N}$ was 1-1. Now, suppose x and y are in the domain of $\cup\mathcal{N}$, and that $x \neq y$, and for some v and w, $\langle x, v\rangle \in \cup\mathcal{N}$ and $\langle y, w\rangle \in \cup\mathcal{N}$. Then, for some functions f and g in $\mathcal{N}$, $\langle x, v\rangle \in f$ and $\langle y, w\rangle \in g$. Without loss of generality, suppose that $f \subseteq g$. Then g is 1-1, meaning $v \neq w$. Hence $\cup\mathcal{N}$ is 1-1.

Exercise 1.2

First, let us note that On is not a set. For if it were, it would then be a transitive set of ordinals, and thus would be an ordinal. Then that would mean $On \in On$, contrary to Theorem 1.8 (of Chapter 5). Thus On is not a set.

Now, let A be the class $On \cup \{\{\{\emptyset\}\}\}$. Thus $x \in A$ iff either x is an ordinal, or $x = \{\{\emptyset\}\}$. Note that $\{\{\emptyset\}\}$ is not an ordinal, as if it were, then it would be transitive, which would imply that $\emptyset \in \{\{\emptyset\}\}$, which is not the case. Hence On is a proper subclass of A.

Then define the relation $x \leq y$ as follows: $x \leq y$ iff x and y are both ordinals and $x \subseteq y$, or $y = \{\{\emptyset\}\}$. Note that this defintion implies that $\{\{\emptyset\}\} \leq y$ only holds if $y = \{\{\emptyset\}\}$. We first show that $\leq$ is a well ordering of A.

O_1: If x is an ordinal, then $x \subseteq x$, so that $x \leq x$ holds. If $x = \{\{\emptyset\}\}$, then $x \leq x$ by definition.

O_2: Suppose $(x \leq y) \wedge (y \leq z)$. If all of x, y and z are ordinals, then obviously $x \leq z$. In fact, the only one of x, y and z which could be $\{\{\emptyset\}\}$ is z. In that case, $x \leq z$ holds by definition.

O_3: Suppose $x \leq y$ and $y \leq x$. Then either $x = y = \{\{\emptyset\}\}$, or both x and y are ordinals, in which case $x \subseteq y$ and $y \subseteq x$, so that $x = y$. Thus $x = y$.

O_4: If x and y are both ordinals, then of course either $x \leq y$ or $y \leq x$. If $y = \{\{\emptyset\}\}$, then $x \leq y$ (similarly with x).

O_5: For any subclass B of A, either $\{\{\emptyset\}\} \in B$ or not. If it is, then, $\{\{\emptyset\}\}$ is the least element of B (as $x \leq \{\{\emptyset\}\}$ holds for all elements in A). If it is not in B, then B is a subclass of On, hence (because On is well ordered under inclusion), B has a least element.

Thus $\leq$ is a well ordering of A. Now, On is a proper lower section of A: It is a proper subclass of A, and $x < \{\{\emptyset\}\}$ for all elements in On. But On is not a set. Thus we have an example of a well ordering which is not a proper well ordering.